國立臺灣大學電機資訊學院電信工程學研究所

碩士論文

Graduate Institute of Communication Engineering

College of Electrical Engineering and Computer Science

National Taiwan University

Master Thesis

# 上行非正交多工系統之接收器設計與效能比較

# Receiver Design and Performance Comparison for Uplink Non-Orthogonal Multiple Access Systems

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中華民國 115 年 6 月

June, 2026

# 國立臺灣大學碩士學位論文

# 口試委員會審定書

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本論文係洪睿帆君（R13942124）在國立臺灣大學電信工程學研究所完成之碩士學位論文，於民國 115 年 6 月 12 日承下列考試委員審查通過及口試及格，特此證明

口試委員：________________________________

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# Acknowledgements

I would like to express my sincere gratitude to my advisor, Professor Mao-Chao Lin, for his guidance, support, and valuable suggestions throughout this research. His advice helped me clarify the research direction and improve the technical depth of this thesis.

I am also grateful to the teachers, colleagues, and laboratory members who provided discussions, comments, and assistance during the development of this work. Their feedback helped me refine the system models, simulation settings, and presentation of the results.

Finally, I would like to thank my family and friends for their continuous encouragement and support. Their patience and understanding made it possible for me to complete this thesis.

# 摘要

本論文研究上行非正交多工（non-orthogonal multiple access, NOMA）系統之多使用者接收技術，並由單一資源、未編碼多資源、編碼多資源、OFDM，以及非同步傳輸等情境逐步比較不同接收器的效能與複雜度取捨。相較於正交多工將使用者分配至彼此正交的時間、頻率或碼域資源，NOMA 允許多個使用者共享相同資源以提升系統負載與頻譜使用效率。然而，上行 NOMA 的可靠度主要取決於基地台接收端是否能有效分離重疊的使用者訊號，因此接收器架構成為系統可行性的關鍵。

首先，本論文在單一資源模型下比較功率域 NOMA 搭配軟式連續干擾消除（soft successive interference cancellation, soft SIC）與增益域多工（gain division multiple access, GDMA）。GDMA 利用複數通道增益的振幅與相位進行聯合偵測，而 PD-NOMA 主要仰賴接收功率差異與 SIC 順序。模擬結果顯示，GDMA 在未編碼位元錯誤率方面通常優於 soft SIC，特別是在使用者接收功率相近或使用者數增加時更為明顯。接著，本論文將 PD-NOMA 延伸至多資源向量觀測模型，並以 MMSE-SIC 接收器與 HDS-CDMA 聯合偵測基準進行比較。多資源觀測可利用跨資源通道向量改善使用者分離能力，但在未編碼情況下，HDS-CDMA 仍因聯合多使用者偵測而具有較佳效能。

為了降低 SIC 錯誤傳播，本論文進一步發展編碼多資源 MMSE-SIC-IDD 接

收器。此接收器結合解碼器輔助之軟式干擾消除、變異數感知 MMSE 濾波，以及偵測器與解碼器之間的外部迭代回饋。模擬結果顯示，導入通道編碼與 IDD 後，MMSE-SIC-IDD 可明顯縮小與 MAP-based 聯合偵測之間的差距，並在較低複雜度下提供具競爭力的 BLER 表現。本論文亦討論 MMSE-PIC-IDD 與 MAP-IDD 作為延遲、複雜度與效能之比較基準。最後，本文將接收器延伸至 coded OFDM NOMA 與非同步 OFDM NOMA，分析頻率選擇性通道、偵測順序、SIC/PIC 架構，以及時間偏移所造成之相位旋轉與額外干擾對接收器穩健性的影響。

**關鍵字**：非正交多工、功率域非正交多工、增益域多工、連續干擾消除、迭代偵測與解碼

# Abstract

This thesis studies multiuser receiver design for uplink non-orthogonal multiple access (NOMA) systems. The comparison progresses from uncoded single-resource detection to uncoded and coded multi-resource reception, and is further extended to coded OFDM and asynchronous OFDM uplink scenarios. In contrast to orthogonal multiple access (OMA), where users are assigned disjoint time, frequency, or code resources, NOMA allows multiple users to share the same resource in order to improve system loading and spectral efficiency. The price of this resource sharing is that the base station must separate superimposed user signals reliably; therefore, the receiver architecture is a decisive factor in uplink NOMA performance.

The thesis first compares single-resource power-domain NOMA (PD-NOMA) with soft successive interference cancellation (soft SIC) and gain division multiple access (GDMA). GDMA exploits both the magnitudes and phases of complex channel gains through joint detection, whereas PD-NOMA mainly relies on received-power disparity and successive cancellation. The results show that GDMA generally achieves better uncoded BER than soft SIC, especially when users have similar received powers or when the number of users increases. The thesis then extends PD-NOMA to a multi-resource vector observation model and evaluates MMSE-SIC against HDS-CDMA joint detection. Multi-resource observations improve user separability by exploiting channel vectors across shared resources, but HDS-CDMA remains the stronger uncoded benchmark because of its joint multiuser detection capability.

To mitigate SIC error propagation, a coded multi-resource MMSE-SIC-IDD receiver is developed. The receiver integrates decoder-aided soft interference cancellation, variance-aware MMSE filtering, and outer detector-decoder feedback. Simulation results show that, after channel coding and IDD are introduced, MMSE-SIC-IDD becomes much more competitive with MAP-based joint detection while maintaining lower complexity. MMSE-PIC-IDD and MAP-IDD are also considered as references for the latency, complexity, and performance tradeoff. Finally, the receiver framework is extended to coded OFDM NOMA and asynchronous OFDM NOMA, where frequency selectivity, detection ordering, SIC/PIC receiver structures, timing-offset-induced phase rotations, and additional MAI/ISI effects are discussed.

**Keywords:** NOMA, PD-NOMA, GDMA, MMSE-SIC, iterative detection and decoding

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# Chapter 1 Introduction

This chapter introduces the background and motivation of uplink non-orthogonal multiple access (NOMA) receiver design. Starting from the demand for higher spectral efficiency and massive connectivity, this chapter explains why intentional resource sharing requires effective multiuser detection, and then summarizes the NOMA schemes, research questions, contributions, and thesis organization considered in this work.

## 1.1 Background and Motivation

Multiple access (MA) techniques determine how multiple users share limited communication resources. In conventional orthogonal multiple access (OMA), users are assigned disjoint resources in time, frequency, code, or subcarrier index. Since users do not overlap on the same resource, inter-user interference (IUI) is avoided and a single-user detector (SUD) is usually sufficient. This simplicity is attractive, but the number of simultaneously supportable users is fundamentally limited by the number of orthogonal resource units.

Future wireless systems require much higher connection density. Massive machine-type communications (mMTC), random access, and low-latency uplink services require the base station (BS) to support many devices that may transmit short packets sporadically. In such scenarios, assigning one orthogonal resource to each active device can be inefficient and may introduce additional scheduling latency. Non-orthogonal multiple access (NOMA) improves resource utilization by allowing multiple users to transmit over the same resource Dai et al. [2018], Islam et al. [2016]. The cost of this improved loading is that the receiver must separate superimposed user signals.

In the uplink, the receiver is therefore the central component of a NOMA system. Unlike the downlink, where the BS can precode or allocate power with relatively complete control, the uplink receiver must separate independently transmitted user signals after they have been distorted by fading, noise, synchronization mismatch, and possibly imperfect channel estimation. The same multiple-access loading can lead to very different performance depending on whether the receiver performs joint detection, linear filtering, successive cancellation, or iterative detection and decoding. This thesis focuses on this receiver-side viewpoint and compares three related NOMA directions:

- GDMA and HDS-CDMA, which exploit complex channel gains or resource-domain signatures and are naturally associated with joint ML or MAP detection;

- single-resource PD-NOMA, which separates users mainly by received-power disparity and is commonly implemented with SIC;

- multi-resource PD-NOMA, which preserves the low-complexity SIC structure while extending the observation from a scalar received signal to a vector across multiple shared resources.

## 1.2 Orthogonal and Non-Orthogonal Multiple Access

An MA resource can be a frequency band, a time slot, a spreading signature, or an OFDM subcarrier. In OMA, each user occupies a distinct orthogonal resource; ideally, no IUI is present. In NOMA, multiple users intentionally share a resource, so a multiuser detector (MUD) is required. The tradeoff is summarized as follows. OMA has low receiver complexity and predictable reliability, but its user capacity is limited by the number of orthogonal resources. NOMA can support more users and reduce access latency, but receiver complexity, error propagation, and synchronization sensitivity become important design issues.

NOMA schemes can be categorized by the domain used for user separation. In power-domain NOMA, users are distinguished by received-power differences and decoded successively Islam et al. [2016]. This approach has low receiver complexity, but its reliability depends strongly on the existence of a clear power gap and on the correctness of the early SIC decisions. In gain-domain schemes such as GDMA, users are distinguished by their complex channel gains, including both magnitude and phase Hsu et al. [2019], Lin et al. [2023b]. This provides a richer separation mechanism, but usually requires joint evaluation of multiuser hypotheses. Code-domain NOMA schemes such as SCMA, LDS-CDMA, and HDS-CDMA use sparse or dense resource-domain signatures Lin et al. [2023a], Nikopour and Baligh [2013]. The multi-resource PD-NOMA model considered in this thesis lies between these categories: it does not impose a designed sparse signature, but it uses the vector channel gains across multiple shared resources to improve separability.

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Figure 1.1: Conceptual comparison between OMA and uplink NOMA. In OMA, users occupy orthogonal resources and a single-user detector is sufficient. In NOMA, users share resources and the BS requires a multiuser detector.

## 1.3 Research Questions

The presentation and conference-paper materials motivate three practical questions.

1. In a single shared resource, how does soft-SIC PD-NOMA compare with GDMA under uncoded BPSK transmission?
2. When the receiver observes multiple shared resources, how much can MMSE-SIC improve PD-NOMA, and how close is it to HDS-CDMA with joint ML or MAP detection?
3. In coded systems, can decoder-aided soft cancellation and outer detector-decoder iterations reduce SIC error propagation enough for MMSE-SIC-IDD to approach coded joint-detection baselines?

The later chapters further extend the comparison to coded OFDM and asynchronous NOMA. These extensions are important because practical uplink systems often use OFDM, frequency-selective channels, cyclic prefixes, and imperfect timing alignment.

## 1.4 Contributions

The main contributions of this thesis draft are:

- A unified single-resource uplink model is used to compare GDMA and PD-NOMA with soft SIC. The comparison highlights the difference between channel-gain-based joint detection and power-domain successive cancellation.

- A multi-resource PD-NOMA model is formulated as a vector observation problem. MMSE-SIC detection with post-SINR ordering and amplitude-based ordering is derived and compared with HDS-CDMA.

- A coded MMSE-SIC-IDD receiver is developed. The receiver combines decoder-aided interference cancellation, variance-aware MMSE filtering, and outer detector-decoder feedback.

- MMSE-PIC-IDD and MAP-IDD are included as coded receiver references, allowing the receiver-complexity and performance tradeoff to be discussed for three, four, and six users.

- The framework is extended to coded OFDM NOMA over quasi-static multipath channels and to asynchronous uplink transmission with timing offsets within the cyclic prefix.

## 1.5 Thesis Organization

Chapter 2 studies single-resource uplink systems, including GDMA and PD-NOMA with soft SIC. This chapter is placed first because it isolates the most basic difference between channel-gain-based joint detection and received-power-based successive cancellation. Chapter 3 extends PD-NOMA to multiple resources and derives the MMSE-SIC receiver. It is included to examine whether vector observations can recover part of the separability lost by scalar PD-NOMA while retaining lower complexity than joint detection. Chapter 4 presents coded multi-resource NOMA with IDD, including MMSE-SIC-IDD, MMSE-PIC-IDD, and MAP-IDD. This chapter addresses the fact that practical receivers operate with channel coding and can use decoder reliability to improve interference cancellation. Chapter 5 extends the coded receiver to OFDM so that frequency-selective channels, cyclic prefixes, and subcarrier-domain detection can be considered. Chapter 6 discusses asynchronous OFDM NOMA, which is necessary because uplink random access rarely provides perfect timing alignment among users. Chapter 7 concludes the thesis and lists future research directions.

# Chapter 2 Single-Resource Uplink Systems

This chapter studies the scalar uplink NOMA model in which all users share one resource. GDMA is reviewed as a joint-detection approach that exploits complex channel gains, while PD-NOMA with soft SIC is considered as a low-complexity receiver that mainly relies on received-power ordering. This comparison establishes the baseline performance gap and identifies the SIC error-propagation issue that motivates the later multi-resource and coded receiver designs.

## 2.1 System Model

Consider an uplink system with $U$ users transmitting simultaneously over one shared resource. The resource may be a frequency band, time slot, signature, or subcarrier. The received signal at the BS is

$$y = \sum_{k=1}^{U} \sqrt{P_k \beta_k} h_k x_k + n, \tag{2.1}$$

where $P_k$ is the transmit power of user $k$, $\beta_k$ is the large-scale fading coefficient, $h_k$ is the small-scale Rayleigh flat-fading coefficient, $x_k$ is the transmitted symbol, $g_k = \sqrt{P_k \beta_k} h_k$

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Figure 2.1: Single-resource uplink NOMA system model with $U$ users sharing one resource.

is the effective channel coefficient, and $n \sim \mathcal{CN}(0, N_0)$ is complex AWGN. The noise variance per real dimension is denoted by $\sigma^2 = N_0/2$. Unless otherwise stated, $P_k = 1$ is assumed. For BPSK modulation, $x_k \in \{+1, -1\}$. Perfect channel state information at the receiver (CSIR) is assumed in the baseline comparisons. The performance metric for uncoded transmission is bit error rate (BER), and $E_b/N_0$ denotes the average transmit energy per information bit per user.

## 2.2 GDMA Detection

GDMA separates users by their complex channel gains. Unlike PD-NOMA, which relies mainly on received-power differences, GDMA uses both the magnitude and phase of $g_k$. Therefore, even when two users have comparable received powers, their superimposed signal points may remain distinguishable if the complex channel coefficients produce sufficiently separated constellation levels. This property makes GDMA an appropriate high-performance reference for the scalar system, although it also introduces a complexity cost because the detector must evaluate the possible multiuser symbol combinations. For BPSK, all possible multiuser symbol combinations form

$$\mathcal{X} = \{\boldsymbol{x}^{(0)}, \boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(2^U-1)}\}, \tag{2.2}$$

where $\boldsymbol{x}^{(\ell)} = [x_1^{(\ell)}, \ldots, x_U^{(\ell)}]^T$ and $x_k^{(\ell)} \in \{+1, -1\}$. The corresponding superimposed signal levels are

$$\begin{aligned} s_\ell &= (-1)^{l_2[1]} \cdot \sqrt{\beta_1} h_1 + (-1)^{l_2[2]} \cdot \sqrt{\beta_2} h_2 + \cdots + (-1)^{l_2[U]} \cdot \sqrt{\beta_U} h_U \\ &= \sum_{k=1}^{U} (-1)^{l_2[k]} \cdot \sqrt{\beta_k} h_k, \quad \ell = 0, 1, \ldots, 2^U - 1 \end{aligned} \tag{2.3}$$

For $m$ coded bits per modulation symbol, the number of levels is $L = 2^{mU}$; for BPSK, $m = 1$.

For the special case of $U = 2$ and BPSK, the mapping from coded bits $c_k$ to BPSK symbols $x_k$ and the resulting superimposed signal levels $s_\ell$ is shown in Table 2.1. Here we use the standard BPSK mapping $x_k = 1 - 2c_k$ for $c_k \in \{0, 1\}$.

Table 2.1: Mapping between coded bits, BPSK symbols, and superimposed signal levels for $m = 1$ and $U = 2$.

| $\ell$ | $c_1$ | $c_2$ | $x_1$ | $x_2$ | $s_\ell$ |
|---|---|---|---|---|---|
| 0 | 0 | 0 | +1 | +1 | $\sqrt{\beta_1}h_1 + \sqrt{\beta_2}h_2$ |
| 1 | 0 | 1 | +1 | -1 | $\sqrt{\beta_1}h_1 - \sqrt{\beta_2}h_2$ |
| 2 | 1 | 0 | -1 | +1 | $-\sqrt{\beta_1}h_1 + \sqrt{\beta_2}h_2$ |
| 3 | 1 | 1 | -1 | -1 | $-\sqrt{\beta_1}h_1 - \sqrt{\beta_2}h_2$ |

Given the received signal $y$, the APP of each possible superimposed level is

$$P(s_\ell | y) = \frac{\exp\left(-\frac{|y - s_\ell|^2}{N_0}\right)}{\sum_{\ell'} \exp\left(-\frac{|y - s_{\ell'}|^2}{N_0}\right)}. \tag{2.4}$$

The LLR of user $k$ is then obtained by marginalizing over all symbol combinations in

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Figure 2.2: Example of the GDMA BPSK mapping table for two users. Each row represents one candidate multiuser symbol vector and its corresponding superimposed signal level.

which $x_k = +1$ or $x_k = -1$:

$$L_k = \log \frac{\sum_{\ell: x_k^{(\ell)}=+1} \exp\left(-\frac{|y-s_\ell|^2}{N_0}\right)}{\sum_{\ell: x_k^{(\ell)}=-1} \exp\left(-\frac{|y-s_\ell|^2}{N_0}\right)}. \tag{2.5}$$

The detected bit is decided according to the sign of $L_k$.

### 2.2.1 Complexity of GDMA

GDMA joint detection has exponential complexity in the number of users because $2^{mU}$ candidate symbol vectors must be evaluated. This cost is acceptable for small $U$, but it becomes challenging in dense random-access scenarios. The advantage is that GDMA does not discard the phase information of the channel gains and can distinguish users even when their received powers are similar, provided that their complex channel coefficients create separable superimposed levels.

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Figure 2.3: GDMA detection flow: enumerate superimposed signal levels, compute APPs, marginalize APPs into user LLRs, and make hard decisions.

## 2.3 PD-NOMA with Soft SIC

PD-NOMA separates users mainly by received-power difference. In the uplink, the BS usually detects and cancels the strongest received signal first, then repeats the same operation on the residual signal. This principle is attractive because the receiver complexity grows much more slowly than exhaustive joint detection. However, the same sequential structure also creates the main weakness of SIC: if an early decision or soft estimate is unreliable, the cancellation residue becomes structured interference for later users. The instantaneous received-power metric of user $k$ is

$$\gamma_k = P_k \beta_k \left| h_k \right|^2 = \left| g_k \right|^2 . \tag{2.6}$$

The SIC order is denoted by

$$\pi = \{\pi(1), \pi(2), \ldots, \pi(U)\}, \tag{2.7}$$

where $\pi(i)$ is the user detected at stage $i$. A common ordering rule is to detect the user with the highest received power first. After ordering, the user indexes can be renumbered so that user $1$ is detected first, user $2$ second, and so on.

At SIC stage $i$, the residual signal is denoted by $y^{(i)}$, with $y^{(1)} = y$. The detector approximates the remaining undetected users as Gaussian interference and computes the LLR of user $i$. An equivalent scalar observation can be written as

$$y^{(i)} = g_i x_i + \eta_i, \tag{2.8}$$

where $\eta_i$ includes AWGN, undetected-user interference, and residual cancellation errors. If $\eta_i$ is approximated as circularly symmetric complex Gaussian with variance $\sigma^2_{\text{eff},i}$, then for a general modulation alphabet $\mathcal{S}$, the LLR of the $q$th bit can be approximated by

$$L_{i,q} \approx \log \frac{\sum_{s\in\mathcal{S}_{q,+}} \exp\left(-\frac{\left|y^{(i)}-g_i s\right|^2}{\sigma^2_{\text{eff},i}}\right)}{\sum_{s\in\mathcal{S}_{q,-}} \exp\left(-\frac{\left|y^{(i)}-g_i s\right|^2}{\sigma^2_{\text{eff},i}}\right)}, \tag{2.9}$$

where $\mathcal{S}_{q,+}$ and $\mathcal{S}_{q,-}$ denote the subsets of constellation symbols whose $q$th bit is mapped to $+1$ and $-1$, respectively. The corresponding soft estimate follows the posterior-mean interpretation used in interference-cancellation receivers Shapiro and Dunyak [2005]:

$$\hat{s}_i = \mathbb{E}[x_i|y^{(i)}] \approx \frac{\sum_{s\in\mathcal{S}} s\exp\left(-\frac{\left|y^{(i)}-g_i s\right|^2}{\sigma^2_{\text{eff},i}}\right)}{\sum_{s\in\mathcal{S}} \exp\left(-\frac{\left|y^{(i)}-g_i s\right|^2}{\sigma^2_{\text{eff},i}}\right)}, \tag{2.10}$$

and the conditional variance is

$$\sigma^2_{\hat{s}_i} = \mathbb{E}\left[\left|x_i - \hat{s}_i\right|^2 \middle| y^{(i)}\right] \approx \frac{\sum_{s\in\mathcal{S}} |s-\hat{s}_i|^2 \exp\left(-\frac{\left|y^{(i)}-g_i s\right|^2}{\sigma^2_{\text{eff},i}}\right)}{\sum_{s\in\mathcal{S}} \exp\left(-\frac{\left|y^{(i)}-g_i s\right|^2}{\sigma^2_{\text{eff},i}}\right)}. \tag{2.11}$$

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Figure 2.4: Soft-SIC receiver flow for single-resource PD-NOMA. Users are detected successively according to received power, and soft symbol estimates are subtracted from the residual signal.

Since the simulations in this chapter use BPSK, $\mathcal{S} = \{+1, -1\}$, and the LLR, soft estimate, and conditional variance reduce to

$$L_i \approx \frac{4\Re\{g_i^* y^{(i)}\}}{\sigma^2_{\text{eff},i}}, \qquad \hat{s}_i = \tanh\left(\frac{L_i}{2}\right), \qquad \sigma^2_{\hat{s}_i} = 1 - \tanh^2\left(\frac{L_i}{2}\right). \tag{2.12}$$

The residual signal for the next stage is updated by

$$y^{(i+1)} = y^{(i)} - g_i \hat{s}_i. \tag{2.13}$$

This process is repeated until all users are detected.

## 2.4 Simulation Results and Discussion

The single-resource simulations use BPSK modulation over independent Rayleigh flat-fading channels. Soft SIC uses instantaneous received-power ordering. The GDMA receiver performs joint detection using the APP/LLR rule in (2.5). The first compari-

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Figure 2.5: BER comparison of GDMA and soft-SIC PD-NOMA for two users with $(\beta_1, \beta_2) = (1, 1)$.

son considers two users with large-scale fading coefficients $(\beta_1, \beta_2) = (1, 1)$, $(1, 0.1)$, and $(1, 0.01)$. The second comparison considers a three-user case with $(\beta_1, \beta_2, \beta_3) = (1, 0.1, 0.01)$.

The main observation is that GDMA consistently outperforms soft SIC. This is expected because GDMA jointly evaluates all possible multiuser symbol combinations, while SIC makes local decisions and is vulnerable to error propagation. When the large-scale fading gap increases, soft SIC becomes more effective because the received-power disparity provides a clearer detection order and reduces the ambiguity of early cancellation stages. Since the received-power metric depends on both the transmit power $P_k$ and the large-scale fading coefficient $\beta_k$, power allocation can partially compensate for path-loss differences and shape the SIC ordering condition. However, relying on such disparity also introduces a fairness concern, which is consistent with the user-pairing fairness issue discussed in Ding et al. [2015].

The three-user result further clarifies the limitation of scalar PD-NOMA. As the number of users increases, soft SIC becomes more susceptible to error propagation because

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Figure 2.6: BER comparison of GDMA and soft-SIC PD-NOMA for two users with $(\beta_1, \beta_2) = (1, 0.1)$.

each cancellation stage depends on the reliability of the previous soft estimates. Residual cancellation errors can accumulate through the successive process and degrade the following detections. In contrast, GDMA evaluates the multiuser signal constellation as a whole and can therefore maintain favorable BER performance as the number of users increases. This performance advantage comes with an important cost: the GDMA hypothesis space grows exponentially with $U$, so the decoding complexity also increases exponentially. The conclusion of this chapter is therefore twofold. First, joint use of complex channel information is valuable for uplink NOMA detection. Second, a practical low-complexity receiver needs additional observations or additional reliability information if it is to narrow the gap to joint detection. These two points motivate the multi-resource vector model in the next chapter.

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Figure 2.7: BER comparison of GDMA and soft-SIC PD-NOMA for two users with $(\beta_1, \beta_2) = (1, 0.01)$.

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Figure 2.8: BER comparison of GDMA and soft-SIC PD-NOMA for three users with $(\beta_1, \beta_2, \beta_3) = (1, 0.1, 0.01)$.

# Chapter 3 Multi-Resource PD-NOMA

This chapter extends single-resource PD-NOMA to a multiple-resource system, where several shared-resource observations are used to improve multiuser separation. Since single-resource PD-NOMA mainly relies on received-power differences, the multiple-resource model allows the BS to further exploit resource-domain channel-vector information. In this chapter, MMSE-SIC is used as the detector, post-SINR ordering and amplitude-based ordering are considered as two detection-ordering methods, and the performance is compared with HDS-CDMA joint detection Lin et al. [2023a].

## 3.1 Vector Observation Model

In multi-resource PD-NOMA, each user may transmit over multiple shared resources, and the BS jointly processes the received vector. This model should be understood as a receiver-side generalization of PD-NOMA rather than a designed spreading scheme: users are not assigned carefully optimized sparse signatures, but their channel responses across the shared resources naturally form distinct vectors. Let $R$ denote the number of resources and $U$ the number of users. At one symbol time, the received vector is

$$\boldsymbol{y} = \boldsymbol{H}\boldsymbol{x} + \boldsymbol{n}, \tag{3.1}$$

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Figure 3.1: Multi-resource PD-NOMA model. Multiple users share multiple resources, and the BS jointly processes the stacked resource-domain observations.

where $\boldsymbol{y} \in \mathbb{C}^{R\times 1}$, $\boldsymbol{x} \in \mathbb{C}^{U\times 1}$, and $\boldsymbol{n} \sim \mathcal{CN}(\mathbf{0}, N_0\boldsymbol{I})$. The channel matrix $\boldsymbol{H} \in \mathbb{C}^{R\times U}$ is

$$\boldsymbol{H} = \begin{bmatrix} \sqrt{P_1\beta_1}h_{1,1} & \sqrt{P_2\beta_2}h_{1,2} & \cdots & \sqrt{P_U\beta_U}h_{1,U} \\ \sqrt{P_1\beta_1}h_{2,1} & \sqrt{P_2\beta_2}h_{2,2} & \cdots & \sqrt{P_U\beta_U}h_{2,U} \\ \vdots & \vdots & \ddots & \vdots \\ \sqrt{P_1\beta_1}h_{R,1} & \sqrt{P_2\beta_2}h_{R,2} & \cdots & \sqrt{P_U\beta_U}h_{R,U} \end{bmatrix}. \tag{3.2}$$

The $k$th column $\boldsymbol{h}_k$ represents the resource-domain channel vector of user $k$.

For the representative two-resource and three-user case, the model is

$$\begin{bmatrix} y_A \\ y_B \end{bmatrix} = \begin{bmatrix} h_{A,1} & h_{A,2} & h_{A,3} \\ h_{B,1} & h_{B,2} & h_{B,3} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} n_A \\ n_B \end{bmatrix}, \tag{3.3}$$

where the large-scale fading and transmit-power factors are absorbed into each $h_{r,k}$.

## 3.2 Relation to Other Multiuser Detection Models

The linear vector model in (3.1) is similar to models used in multi-antenna reception and PDMA Endo et al. [2012], Kong et al. [2016], Li [2013]. In multi-antenna systems, user separation is achieved through spatial-domain observations. In PDMA, separation is further assisted by user-specific pattern mapping across resources. The multi-resource PD-NOMA model considered here does not rely on sparse signatures or pattern design Mohammadkarimi et al. [2018]. Instead, user separation comes from the natural channel vectors across shared resources. It is therefore best interpreted as a resource-domain extension of PD-NOMA rather than a full code-domain NOMA design.

This interpretation is important for the receiver comparison. If a system uses designed signatures, performance can depend strongly on the signature matrix and factor-graph structure. In the present model, the comparison focuses on the receiver's ability to exploit the channel matrix itself. HDS-CDMA is consequently used as a joint-detection baseline, whereas MMSE-SIC is used to represent a lower-complexity detector that processes the same vector observation through linear filtering and successive cancellation.

## 3.3 HDS-CDMA Joint Detection Baseline

HDS-CDMA is used as a joint-detection baseline for multi-resource systems Lin et al. [2023a]. Its role in this chapter is analogous to GDMA in the single-resource chapter: it provides a reference for the performance that can be obtained when the receiver jointly evaluates the multiuser hypotheses over all resources. For BPSK, define the candidate symbol-vector set

$$\mathcal{X} = \{\boldsymbol{x}^{(0)}, \boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(2^U-1)}\}. \tag{3.4}$$

For a candidate vector $\boldsymbol{x}^{(\ell)}$, the likelihood is

$$p(\boldsymbol{y}|\boldsymbol{x}^{(\ell)}) \propto \exp\left(-\frac{\left\|\boldsymbol{y} - \boldsymbol{H}\boldsymbol{x}^{(\ell)}\right\|^2}{N_0}\right). \tag{3.5}$$

The ML detector selects the most likely vector, while the MAP detector computes bit-level APPs and LLRs by marginalizing over the candidates. The LLR of user $k$ is

$$L_k = \log \frac{\sum_{\ell: x_k^{(\ell)}=+1} \exp\left(-\frac{\left\|\boldsymbol{y}-\boldsymbol{H}\boldsymbol{x}^{(\ell)}\right\|^2}{N_0}\right)}{\sum_{\ell: x_k^{(\ell)}=-1} \exp\left(-\frac{\left\|\boldsymbol{y}-\boldsymbol{H}\boldsymbol{x}^{(\ell)}\right\|^2}{N_0}\right)}. \tag{3.6}$$

This detector provides an optimal or near-optimal error-performance bound for the considered multi-resource NOMA system, but its complexity grows exponentially with $U$.

## 3.4 MMSE-SIC Detection

The MMSE-SIC receiver reduces complexity by detecting users successively after linear MMSE filtering. The filter first suppresses the interference components that are linearly separable from the desired user's channel vector, and the SIC stage then removes the detected user's contribution from the residual vector. Compared with single-resource soft SIC, this receiver can exploit both magnitude and phase differences across resources. Compared with joint ML or MAP detection, it avoids enumerating all $2^U$ candidate vectors. The main quantities needed after filtering are the desired effective channel gain, the residual interference-plus-noise variance, the LLR, and the soft estimate used for cancellation. To detect user $k$, the MMSE filter is

$$\boldsymbol{w}_k^H = \boldsymbol{h}_k^H \left(\boldsymbol{H}\boldsymbol{H}^H + N_0\boldsymbol{I}\right)^{-1}. \tag{3.7}$$

The filter output is

$$z_k = \boldsymbol{w}_k^H \boldsymbol{y} = \boldsymbol{w}_k^H \boldsymbol{h}_k x_k + \sum_{j \neq k} \boldsymbol{w}_k^H \boldsymbol{h}_j x_j + \boldsymbol{w}_k^H \boldsymbol{n}. \tag{3.8}$$

The effective channel gain is

$$a_k = \boldsymbol{w}_k^H \boldsymbol{h}_k, \tag{3.9}$$

and the effective residual interference-plus-noise variance is

$$\sigma_{k,\text{eff}}^2 = \sum_{j \neq k} \left|\boldsymbol{w}_k^H \boldsymbol{h}_j\right|^2 + N_0 \left\|\boldsymbol{w}_k\right\|^2. \tag{3.10}$$

### 3.4.1 Detection Ordering

Two ordering strategies are considered. The first is post-SINR ordering. The post-filtering SINR of user $k$ is

$$\Gamma_k^{\text{post}} = \frac{\left|\boldsymbol{w}_k^H \boldsymbol{h}_k\right|^2}{\sum_{j \neq k} \left|\boldsymbol{w}_k^H \boldsymbol{h}_j\right|^2 + N_0 \left\|\boldsymbol{w}_k\right\|^2}. \tag{3.11}$$

The receiver detects the user with the largest post-SINR first. This criterion uses both the magnitude and phase of the channel matrix and reflects the actual quality after filtering.

The second strategy is amplitude-based ordering, where users are sorted by a channel magnitude or received-power metric such as

$$\Gamma_k^{\text{amp}} = \|\boldsymbol{h}_k\|^2 . \tag{3.12}$$

This ordering is simpler because it uses only channel magnitudes, but it ignores how the MMSE filter suppresses multiuser interference. The simulations show that amplitude-based ordering can degrade significantly in equal-large-scale-fading conditions.

### 3.4.2 SIC Procedure

After the order is chosen, users are renumbered as $1, 2, \ldots, U$. At stage $i$, the residual vector is $\boldsymbol{y}^{(i)}$ and the remaining channel matrix is

$$\boldsymbol{H}^{(i)} = [\boldsymbol{h}_i, \boldsymbol{h}_{i+1}, \ldots, \boldsymbol{h}_U]. \tag{3.13}$$

The MMSE filter is recomputed using the reduced matrix. The filter output is

$$z_i = \boldsymbol{w}_i^H \boldsymbol{y}^{(i)} = \boldsymbol{w}_i^H \boldsymbol{h}_i x_i + \sum_{j>i} \boldsymbol{w}_i^H \boldsymbol{h}_j x_j + \boldsymbol{w}_i^H \boldsymbol{n}. \tag{3.14}$$

The effective variance is

$$\sigma_{i,\text{eff}}^2 = \sum_{j>i} \left| \boldsymbol{w}_i^H \boldsymbol{h}_j \right|^2 + N_0 \|\boldsymbol{w}_i\|^2 . \tag{3.15}$$

After MMSE filtering, the output can be approximated as an equivalent scalar channel

$$z_i = a_i x_i + \eta_i, \qquad a_i = \boldsymbol{w}_i^H \boldsymbol{h}_i, \qquad \eta_i \sim \mathcal{CN}(0, \sigma_{i,\text{eff}}^2). \tag{3.16}$$

For a general bit-labeled modulation alphabet $\mathcal{S}$, let $b_q(s) \in \{0,1\}$ denote the $q$th label bit of symbol $s$, and define

$$\mathcal{S}_{q,b} = \{s \in \mathcal{S} : b_q(s) = b\}, \qquad b \in \{0,1\}. \tag{3.17}$$

In this chapter, the transmitted coded bits are assumed to be independent and equally likely. Therefore, each constellation symbol has the same prior probability, and the prior term cancels in the likelihood ratio. Under the scalar Gaussian approximation in (3.16), the standard likelihood-marginalization form of the soft-output demapper Shapiro and Dunyak [2005] gives the detector LLR of the $q$th bit as

$$L_{i,q} \approx \log \frac{\sum_{s \in \mathcal{S}_{q,0}} \exp\left(-\frac{|z_i - a_i s|^2}{\sigma_{i,\text{eff}}^2}\right)}{\sum_{s \in \mathcal{S}_{q,1}} \exp\left(-\frac{|z_i - a_i s|^2}{\sigma_{i,\text{eff}}^2}\right)}, \tag{3.18}$$

where the numerator marginalizes over all constellation points whose $q$th bit is zero and the denominator marginalizes over those whose $q$th bit is one. Since the simulations in this chapter use BPSK, $\mathcal{S} = \{+1, -1\}$ with $0 \mapsto +1$ and $1 \mapsto -1$, and (3.18) becomes

$$L_i \approx \frac{4\Re\{a_i^* z_i\}}{\sigma_{i,\text{eff}}^2}. \tag{3.19}$$

The corresponding soft symbol estimate is the posterior mean obtained from the same equally likely symbol prior:

$$\hat{s}_i = \mathbb{E}[x_i | z_i] \approx \frac{\sum_{s \in \mathcal{S}} s \exp\left(-\frac{|z_i - a_i s|^2}{\sigma_{i,\text{eff}}^2}\right)}{\sum_{s \in \mathcal{S}} \exp\left(-\frac{|z_i - a_i s|^2}{\sigma_{i,\text{eff}}^2}\right)}. \tag{3.20}$$

For BPSK, this posterior mean reduces to $\hat{s}_i = \tanh(L_i/2)$. The expression in (3.20) is therefore a detector-output soft estimate for the uncoded receiver when all transmitted bits

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`figures/mmse_sic_procedure.pdf`

Figure 3.2: MMSE-SIC detection procedure for multi-resource PD-NOMA: ordering, MMSE filtering, LLR calculation, soft estimation, and residual update.

are equally likely. The soft estimate is then subtracted:

$$\boldsymbol{y}^{(i+1)} = \boldsymbol{y}^{(i)} - \boldsymbol{h}_i \hat{s}_i. \tag{3.21}$$

If $\hat{s}_i$ is unreliable, the subtraction leaves a cancellation residual in $\boldsymbol{y}^{(i+1)}$. This residual is one of the main sources of error propagation in MMSE-SIC, and it motivates the variance-aware coded receiver developed in Chapter 4.

## 3.5 Imperfect CSI Model

The baseline simulations assume perfect CSIR, but the presentation also introduces an imperfect CSI model similar to the one used in uplink NOMA ordered decoding studies Gao et al. [2018]. The estimated channel can be written as

$$\hat{\boldsymbol{H}} = \boldsymbol{H} + \boldsymbol{E}, \tag{3.22}$$

where $\boldsymbol{E}$ is the channel-estimation error. A common assumption is that the estimation error is uncorrelated with the channel estimate,

$$\mathbb{E}[\hat{\boldsymbol{H}}\boldsymbol{E}^H] = \mathbf{0}. \tag{3.23}$$

Under this model, ordering and filtering are computed from $\hat{\boldsymbol{H}}$, while residual interference increases because the cancellation vector does not perfectly match the true channel. This issue is especially important for SIC receivers because early-stage cancellation errors propagate to later stages.

## 3.6 Uncoded Simulation Results

The main uncoded multi-resource setting follows the simulation setup in the conference paper and presentation: BPSK transmission, $R = 2$ resources, $U = 3$ users, independent Rayleigh flat fading, and perfect CSI. The detector candidates are HDS-CDMA with ML detection and MMSE-SIC with post-SINR ordering or amplitude-based ordering. Two large-scale fading cases are used:

$$(\beta_1, \beta_2, \beta_3) = (1, 0.1, 0.01) \tag{3.24}$$

and

$$(\beta_1, \beta_2, \beta_3) = (1, 1, 1). \tag{3.25}$$

The average transmit energy per bit per user is normalized by the number of resources.

The results in Fig. 3.3 and Fig. 3.4 show that HDS-CDMA with ML joint detection achieves better uncoded BER than MMSE-SIC in both fading cases. For the unequal large-

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Figure 3.3: Uncoded BER comparison of MMSE-SIC and HDS-CDMA with ML detection for $R = 2$, $U = 3$, and $(\beta_1, \beta_2, \beta_3) = (1, 0.1, 0.01)$.

scale fading case $(1, 0.1, 0.01)$, HDS-CDMA performs about $3$ dB better than MMSE-SIC at BER $= 10^{-4}$ for user 1. For the equal large-scale fading case $(1, 1, 1)$, HDS-CDMA performs about 7 dB better than MMSE-SIC at BER $= 10^{-4}$. Therefore, the performance loss of MMSE-SIC is much more significant when the users have similar average received powers.

The smaller gap in the unequal-fading case is caused by the large-scale fading disparity. Since the users have clearly separated average received powers, the first SIC stage is more reliable and the cancellation error is less likely to dominate the following stages. In particular, for the user with the smallest average received power, MMSE-SIC is close to HDS-CDMA within the simulated SNR range. However, HDS-CDMA still provides the best overall BER performance because ML joint detection evaluates the multiuser symbol hypotheses jointly instead of making sequential hard or soft cancellation decisions.

The ordering comparison further explains the behavior of MMSE-SIC. In the unequal-fading case, amplitude-based ordering only causes a slight degradation around the high-SNR region, for example around $25$ dB for user 1. In the equal-fading case, however,

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Figure 3.4: Uncoded BER comparison of MMSE-SIC and HDS-CDMA with ML detection for $R = 2, U = 3$, and $(\beta_1, \beta_2, \beta_3) = (1, 1, 1)$.

amplitude-based ordering degrades heavily because the users do not have a clear large-scale power separation. Post-SINR ordering is more robust because it measures the quality after MMSE filtering, including residual multiuser interference and noise enhancement, whereas amplitude-based ordering only uses the channel magnitude before filtering. This confirms that large-scale fading disparity provides a clearer detection order for SIC, while equal-power users make SIC more sensitive to ordering errors and error propagation.

These results indicate that multi-resource PD-NOMA improves user separation compared with single-resource soft SIC, but joint detection remains the stronger uncoded benchmark. The remaining performance gap motivates the next chapter. Rather than increasing receiver complexity to full joint detection, Chapter 4 introduces channel coding and IDD so that decoder reliability can be used to reduce SIC error propagation.

# Chapter 4 Coded Multi-Resource Systems with IDD

This chapter introduces channel coding and iterative detection and decoding (IDD) into the multi-resource PD-NOMA receiver. Motivated by the error-propagation problem observed in uncoded SIC receivers, the detector exchanges soft information with the FEC decoder and refines interference cancellation through iterations based on updated a priori information. This chapter develops MMSE-SIC-IDD and compares it with MMSE-PIC-IDD and MAP-IDD to clarify the performance-complexity tradeoff in coded multi-resource uplink NOMA.

## 4.1 Brief introduction of IDD structure

For coded systems, the detector is typically embedded in an iterative detection and decoding (IDD) receiver. In this structure, soft information is exchanged between the detector and the FEC decoder Li et al. [2024], Lin et al. [2024], Sahin et al. [2024]. The detector produces bit-level LLRs for the coded bits and passes them to the FEC decoder, while the decoder uses the code constraints to refine the reliability information and returns updated soft information to the detector. This detector-decoder exchange follows the IDD

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Figure 4.1: General IDD receiver structure. The detector and FEC decoder exchange soft information through iterations based on a priori updates.

structure shown in Fig. 4.1.

The iteration based on a priori update, denoted by $t$, is performed after all users have been decoded. The decoder outputs of all users are used to update their corresponding a priori information, and these updated a priori LLRs are then fed back to the detector. Through the a priori update iteration $t$, the updated a priori information is especially beneficial for both MMSE-SIC and MMSE-PIC. For MMSE-SIC, the updated information helps the stage-by-stage SIC process together with decoded-user feedback. For MMSE-PIC, it is even more critical because parallel cancellation relies directly on the a priori-based soft means of all interfering users.

## 4.2 Coded system model

Let each user transmit a coded bit block

$$\boldsymbol{c}_i = [c_{i,1}, c_{i,2}, \ldots, c_{i,N_{c,i}}]^T. \tag{4.1}$$

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Figure 4.2: MMSE-SIC-IDD receiver. At each SIC stage, a user is detected, decoded, and softly cancelled. After all users are processed, decoder extrinsic information is used for the next a priori update.

After modulation, the coded bits are grouped into $N_s$ modulation symbols. At symbol time $\ell$, the transmitted vector is

$$\boldsymbol{x}_\ell = [s_{1,\ell}, s_{2,\ell}, \ldots, s_{U,\ell}]^T, \tag{4.2}$$

and the received vector is

$$\boldsymbol{y}_\ell = \boldsymbol{H}_\ell \boldsymbol{x}_\ell + \boldsymbol{n}_\ell. \tag{4.3}$$

The received block is $\boldsymbol{Y} = [\boldsymbol{y}_1, \boldsymbol{y}_2, \ldots, \boldsymbol{y}_{N_s}]$. In the block-fading simulations, $\boldsymbol{H}_1 = \boldsymbol{H}_2 = \cdots = \boldsymbol{H}_{N_s}$.

## 4.3 MMSE-SIC-IDD

### 4.3.1 Receiver Procedure

In this section, we introduce the procedure of the MMSE-SIC-IDD receiver. The key feature of the structure is that interference cancellation in the SIC process is performed after channel decoding rather than immediately after detection.

The received coded block is written as

$$\boldsymbol{Y} = [\boldsymbol{y}_1, \boldsymbol{y}_2, \ldots, \boldsymbol{y}_{N_s}], \tag{4.4}$$

where $N_s$ is the number of received coded symbols in the block. For the BPSK coded system, let $\boldsymbol{c}_i = [c_{i,1}, c_{i,2}, \ldots, c_{i,N_s}]^T$ denote the transmitted coded-symbol sequence of user $i$, let $\boldsymbol{H}_\ell$ denote the $\ell$th multiple-access channel, and let $\boldsymbol{x}_\ell = [c_{1,\ell}, c_{2,\ell}, \ldots, c_{U,\ell}]^T$ denote the transmitted signal vector at time index $\ell$. The $\ell$th column of $\boldsymbol{Y}$ is

$$\boldsymbol{y}_\ell = \boldsymbol{H}_\ell \boldsymbol{x}_\ell + \boldsymbol{n}_\ell. \tag{4.5}$$

For higher-order modulation, $c_{i,\ell}$ is replaced by the modulation symbol $s_{i,\ell}$, while the same IDD structure is used.

Without loss of generality, we assume that the detection order has been determined according to the ordering criterion in (3.11), and that the users are re-numbered from 1 to $U$ according to this order. After re-numbering,

$$\Gamma_1^{\text{post}} \geq \Gamma_2^{\text{post}} \geq \cdots \geq \Gamma_U^{\text{post}}, \tag{4.6}$$

and user $i$ is detected at the $i$th SIC stage. For user $i$, the MMSE-SIC detector first computes the bit-level LLRs $\boldsymbol{L}_{\boldsymbol{c}_i}$ using (3.18). The detector also produces the coded soft-estimate vector

$$\hat{\boldsymbol{s}}_i = [\hat{s}_{i,1}, \hat{s}_{i,2}, \ldots, \hat{s}_{i,N_s}]^T. \tag{4.7}$$

The LLRs $\boldsymbol{L}_{\boldsymbol{c}_i}$ are passed to the FEC decoder of user $i$. After decoding, the decoder produces the a posteriori LLRs $\boldsymbol{L}^d_{\boldsymbol{c}_i}$, which are used to generate the decoded soft symbol estimate $\hat{\boldsymbol{s}}^d_i$ according to the posterior-mean principle in (3.20). This decoded soft symbol estimate is then fed back to the MMSE-SIC detector for the next SIC stage, i.e., user $i+1$, where it is used to subtract the interference caused by user $i$. Intuitively, performing cancellation after decoding can improve the reliability of the cancelled signal and thus reduce error propagation in the SIC process.

Due to the successive decoding procedure, the extrinsic information of each user is collected asynchronously. Specifically, after decoding user $i$, the extrinsic LLRs are obtained as

$$\boldsymbol{L}^e_{\boldsymbol{c}_i} = \boldsymbol{L}^d_{\boldsymbol{c}_i} - \boldsymbol{L}_{\boldsymbol{c}_i} \tag{4.8}$$

for each user. A variance matrix is then used to collect the reliability information from all users. Once the extrinsic information of all users has been collected, it is used to update the a priori information for the next IDD iteration. In this way, the decoded extrinsic information of all users is provided as a priori information to the MMSE-SIC detector in the following IDD iteration.

### 4.3.2 Soft Information and Variance

Let $\mathcal{S}_i$ denote the modulation alphabet of user $i$, and let $M_i = \log_2 |\mathcal{S}_i|$ be the number of coded bits per modulation symbol. For a symbol $s \in \mathcal{S}_i$, its binary label is written as

$$\boldsymbol{b}_i(s) = [b_{i,0}(s), b_{i,1}(s), \ldots, b_{i,M_i-1}(s)]. \tag{4.9}$$

Let $L_{i,\ell,m}^{a,(t)}$ denote the a priori LLR of the $m$th coded bit in the $\ell$th modulation symbol of user $i$ at a priori update iteration $t$, with

$$L_{i,\ell,m}^{a,(t)} = \log \frac{P^{a,(t)}(b_{i,\ell,m} = 0)}{P^{a,(t)}(b_{i,\ell,m} = 1)}, \qquad L_{i,\ell,m}^{a,(0)} = 0. \tag{4.10}$$

The corresponding bit probability is

$$P^{a,(t)}(b_{i,\ell,m} = b) = \frac{1}{1 + \exp\left[-(1-2b)L_{i,\ell,m}^{a,(t)}\right]}, \qquad b \in \{0, 1\}. \tag{4.11}$$

Assuming the interleaved coded bits are represented by independent bit priors at the demapper input, the a priori symbol probability is

$$P_{i,\ell}^{a,(t)}(s) = \prod_{m=0}^{M_i-1} P^{a,(t)}(b_{i,\ell,m} = b_{i,m}(s)). \tag{4.12}$$

The soft symbol mean used for a priori update before detection is calculated according to the a priori symbol probability, that is,

$$\mu_{i,\ell}^{(t)} = \mathbb{E}[s_{i,\ell} \mid \boldsymbol{L}_{i,\ell}^{a,(t)}] = \frac{\sum\limits_{s \in \mathcal{S}_i} s P_{i,\ell}^{a,(t)}(s)}{\sum\limits_{s \in \mathcal{S}_i} P_{i,\ell}^{a,(t)}(s)}, \tag{4.13}$$

and the corresponding symbol variance is

$$\sigma_{i,\ell}^{2\,(t)} = \frac{\sum\limits_{s \in \mathcal{S}_i} |s|^2 P_{i,\ell}^{a,(t)}(s)}{\sum\limits_{s \in \mathcal{S}_i} P_{i,\ell}^{a,(t)}(s)} - \left|\mu_{i,\ell}^{(t)}\right|^2. \tag{4.14}$$

For BPSK with $0 \mapsto +1$ and $1 \mapsto -1$, (4.13) and (4.14) reduce to

$$\mu_{i,\ell}^{(t)} = \tanh\left(\frac{L_{i,\ell,0}^{a,(t)}}{2}\right), \qquad \sigma_{i,\ell}^{2\,(t)} = 1 - \left|\mu_{i,\ell}^{(t)}\right|^2. \tag{4.15}$$

The average variance of user $i$ over the coded block is

$$\bar{v}_i^{(t)} = \frac{1}{N_s} \sum_{\ell=1}^{N_s} \sigma_{i,\ell}^{2\,(t)}, \tag{4.16}$$

where $N_s$ is the number of modulation symbols in the coded block. The variance matrix is defined as

$$\boldsymbol{V} = \mathrm{diag}(\bar{v}_1^{(t)}, \bar{v}_2^{(t)}, \ldots, \bar{v}_U^{(t)}), \qquad \bar{v}_i^{(t)} > 0. \tag{4.17}$$

When decoder feedback is reliable, $\bar{v}_i^{(t)}$ becomes small and the receiver treats user $i$ as easier to cancel. When decoder feedback is uncertain, the variance remains large and the receiver avoids overconfident cancellation. This reliability weighting is essential in an IDD receiver.

### 4.3.3 Variance-Aware MMSE Filtering

When all users' extrinsic information from the FEC decoder has been collected, the block-averaged variance of each user is placed in the variance matrix $\boldsymbol{V}$ in (4.17). The variance-aware MMSE filter used in the coded receiver is

$$\tilde{\boldsymbol{w}}_k^H = \bar{v}_k^{(t)} \boldsymbol{h}_k^H \left( \boldsymbol{H} \boldsymbol{V} \boldsymbol{H}^H + 2\sigma^2 \boldsymbol{I} \right)^{-1}. \tag{4.18}$$

Compared with (3.7), the covariance matrix includes $\boldsymbol{V}$, so users with reliable a priori information contribute less uncertainty to the MMSE filtering process.

### 4.3.4 Decoder-Aided SIC Cancellation

To simplify the expression in the following equations, we omit the subscript $\ell$, which denotes that the process is dealing with $\boldsymbol{y}_\ell$. The procedure then goes back to MMSE-SIC stage 1. User 1 benefits from the a priori information of users $2, \ldots, U$ by using

$$\boldsymbol{y}_{(1)}^{(t)} = \boldsymbol{y} - \sum_{j=2}^{U} \boldsymbol{h}_j \mu_j^{(t)}. \tag{4.19}$$

At SIC stage $i > 1$, the detection of user $i$ benefits from the FEC outputs of users $1, 2, \ldots, i-1$ and from the a priori information of users $i+1, \ldots, U$. The residual vector for detecting user $i$ is constructed as

$$\boldsymbol{y}_{(i)}^{(t)} = \boldsymbol{y} - \sum_{j=1}^{i-1} \boldsymbol{h}_j \hat{s}_j^d - \sum_{j=i+1}^{U} \boldsymbol{h}_j \mu_j^{(t)}. \tag{4.20}$$

The first summation cancels the users that have already been decoded in the current SIC pass by using the decoder-aided soft symbol estimates $\hat{s}_j^d$. The second summation cancels the users that have not yet been decoded in the current pass by using the a priori soft symbol means $\mu_j^{(t)}$. This is the coded extension of the uncoded subtraction in Chapter 3.

The cancellation term in (4.19) differs from that used in the uncoded MMSE-SIC detection in Chapter 3. In the uncoded case, the cancelled signal is directly generated from the detector output at the current SIC stage, and the corresponding residual uncertainty is not explicitly incorporated into the subsequent MMSE filtering. In the coded IDD receiver, however, users $1, \ldots, i-1$ have already been decoded in the current SIC pass. Their decoder outputs provide more reliable soft symbol estimates as well as reliability information for quantifying the remaining uncertainty after cancellation. Therefore, the decoder-aided cancellation is modeled together with its residual variance $v_c^{\text{can}}$, whose detailed calculation is given in Section **??**. Similar to the uncoded MMSE-SIC procedure, the downsized channel matrix and the downsized variance matrix are used after each stage. The filter output for user $i$ is

$$z_i^{(t)} = \tilde{\boldsymbol{w}}_i^H \boldsymbol{y}_{(i)}^{(t)}, \tag{4.21}$$

with equivalent scalar gain

$$a_i^{(t)} = \tilde{\boldsymbol{w}}_i^H \boldsymbol{h}_i. \tag{4.22}$$

The effective variance of the residual interference and noise is

$$\sigma_{i,\text{eff}}^{2\ (t)} = \sum_{j>i} \bar{v}_j^{(t)} \left|\tilde{\boldsymbol{w}}_i^H \boldsymbol{h}_j\right|^2 + \sum_{c<i} v_c^{\text{can}} \left|\tilde{\boldsymbol{w}}_i^H \boldsymbol{h}_c\right|^2 + 2\sigma^2 \left\|\tilde{\boldsymbol{w}}_i^H\right\|^2. \tag{4.23}$$

The first term in (4.23) comes from undecoded users whose interference is represented by the a priori variance $\bar{v}_j^{(t)}$. The second term is the cancellation residual caused by the already decoded users. The last term is the filtered noise variance. Therefore, unlike the uncoded MMSE-SIC detection, the coded receiver explicitly accounts for the residual uncertainty of decoder-aided cancellation.

Under the scalar approximation,

$$z_i^{(t)} = a_i^{(t)} x_i + \eta_i^{(t)}, \qquad \eta_i^{(t)} \sim \mathcal{CN}(0, \sigma_{i,\text{eff}}^{2\ (t)}). \tag{4.24}$$

### 4.3.5 LLR calculation and soft estimate

For the $q$th bit of user $i$, define

$$\mathcal{S}_{i,q,b} = \{s \in \mathcal{S}_i : b_{i,q}(s) = b\}, \qquad b \in \{0,1\}. \tag{4.25}$$

After MMSE filtering and decoder-aided cancellation, the detector observes the scalar equivalent channel in (4.23). The detector LLR is therefore computed from $z_i^{(t)}$, the scalar gain $a_i^{(t)}$, and the effective variance $\sigma_{i,\text{eff}}^{2\ (t)}$. The symbol probability induced by the a priori LLRs in the $t$th a priori update is

$$P_i^{a,(t)}(x) = \prod_{m=0}^{M_i-1} P^{a,(t)}\left(b_{i,m} = b_{i,m}(x)\right), \qquad x \in \mathcal{S}_i, \tag{4.26}$$

where the bit probabilities are obtained from $L_{i,\ell,m}^{a,(t)}$ by the LLR-to-probability rule in (4.11). The subscript $\ell$ is omitted in the following scalar detection equations for compactness. The detector produces an extrinsic LLR by using the a priori information of the other bits in the same modulation symbol, but not the a priori information of the target bit itself:

$$L_{c_{i,q}}^e = \log \frac{\sum\limits_{x\in\mathcal{S}_{i,q,0}} \exp\left(-\frac{\left|z_i^{(t)} - a_i^{(t)}x\right|^2}{\sigma_{i,\text{eff}}^{2\ (t)}} + \sum\limits_{\substack{m=0\\m\neq q}}^{M_i-1} \log P^{a,(t)}(b_{i,m} = b_{i,m}(x))\right)}{\sum\limits_{x\in\mathcal{S}_{i,q,1}} \exp\left(-\frac{\left|z_i^{(t)} - a_i^{(t)}x\right|^2}{\sigma_{i,\text{eff}}^{2\ (t)}} + \sum\limits_{\substack{m=0\\m\neq q}}^{M_i-1} \log P^{a,(t)}(b_{i,m} = b_{i,m}(x))\right)}. \tag{4.27}$$

The detector LLR passed to the FEC decoder is obtained by adding the target-bit a priori LLR:

$$L_{c_{i,q}} = \alpha L_{c_{i,q}}^e + L_{i,\ell,q}^{a,(t)}, \tag{4.28}$$

where $\alpha \leq 1$ is an optional damping factor. The undamped case uses $\alpha = 1$, while the implementation used in the simulations may choose a smaller value to avoid overconfident detector feedback.

For BPSK modulation, (4.27) reduces to the scalar LLR expression

$$L_{c_i} = \frac{4\Re\{(a_i^{(t)})^* z_i^{(t)}\}}{\sigma_{i,\text{eff}}^{2\ (t)}}. \tag{4.29}$$

The FEC decoder produces posterior LLRs $\boldsymbol{L}^d_{\boldsymbol{c}_i}$. These posterior LLRs are used to form the decoder-aided soft estimate for SIC cancellation. For the general bit-labeled modulation case, define the decoder posterior symbol probability as

$$P_i^d(x) = \prod_{m=0}^{M_i-1} P^d(b_{i,m} = b_{i,m}(x)), \qquad x \in \mathcal{S}_i, \tag{4.30}$$

where the bit probabilities are obtained from the decoder posterior LLRs using the same LLR-to-probability rule as in (4.11). The decoder-aided posterior soft estimate is

$$\hat{s}_i^d = \mathbb{E}[x_i \mid \boldsymbol{L}^d_{\boldsymbol{c}_i}] = \frac{\sum\limits_{x\in\mathcal{S}_i} xP_i^d(x)}{\sum\limits_{x\in\mathcal{S}_i} P_i^d(x)}, \tag{4.31}$$

with residual variance

$$v_i^{\text{can}} = \frac{\sum\limits_{x\in\mathcal{S}_i} |x|^2 P_i^d(x)}{\sum\limits_{x\in\mathcal{S}_i} P_i^d(x)} - \left|\hat{s}_i^d\right|^2. \tag{4.32}$$

For BPSK modulation, the decoder posterior LLR gives

$$L^d_{c_i} = \text{decoder}(L_{c_i}), \qquad \hat{s}_i^d = \tanh\left(\frac{L^d_{c_i}}{2}\right), \qquad v_i^{\text{can}} = 1 - \left(\hat{s}_i^d\right)^2. \tag{4.33}$$

The quantity $v_i^{\text{can}}$ is the cancellation residual variance used in (4.23). It measures the remaining uncertainty after subtracting the decoder-aided soft estimate $\hat{s}_i^d$ from the received vector. A more reliable decoder output gives a smaller $v_i^{\text{can}}$, which reduces the residual interference term in the following SIC stages.

After all users are processed, the detector a priori LLRs for the next a priori update are obtained from decoder extrinsic information:

$$L^{a,(t+1)}_{i,\ell,q} = L^d_{c_{i,q}} - L_{c_{i,q}}. \tag{4.34}$$

**Algorithm 1** MMSE-SIC-IDD receiver for one coded block

Initialize the a priori LLRs $L^{a,(0)}_{i,\ell,m} = 0$ for all users, coded-symbol positions, and bit positions.
**for** $t = 0, 1, \ldots, t_{\text{max}} - 1$ **do**
  Compute $\mu^{(t)}_{i,\ell}$ and $\sigma^{2\,(t)}_{i,\ell}$ for all users.
  Form the variance matrix $\boldsymbol{V}$ from the block-averaged variances.
  Determine the SIC order using post-SINR ordering and re-number the users as $1, \ldots, U$.
  **for** $i = 1, 2, \ldots, U$ **do**
    Compute the variance-aware MMSE filter in (4.18).
    Build $\boldsymbol{y}^{(t)}_{(i)}$ using (4.19) or (4.20).
    Generate detector extrinsic LLRs using (4.27).
    Decode user $i$ and obtain $\hat{s}^d_i$ and $v^{\text{can}}_i$.
  **end for**
  Set the next a priori LLRs using (4.34).
**end for**
Make hard decisions from the final decoder outputs.

Equivalently, in vector form, $\boldsymbol{L}^e_{\boldsymbol{c}_i} = \boldsymbol{L}^d_{\boldsymbol{c}_i} - \boldsymbol{L}_{\boldsymbol{c}_i}$. The collection of all users' extrinsic information is fed back to the detector as the a priori information for the next update. This a priori update is performed after all users have been decoded.

### 4.3.6 Algorithm Summary

## 4.4 MMSE-PIC-IDD

Parallel interference cancellation (PIC) cancels all other users in parallel instead of successively decoding one user at a time Studer et al. [2011]. For user $k$, the residual vector is

$$\boldsymbol{y}^{(t)}_{k,\ell} = \boldsymbol{y}_\ell - \sum_{j \neq k} \boldsymbol{h}_j \mu^{(t)}_{j,\ell}. \tag{4.35}$$

The variance-aware MMSE filter follows the same form as (4.18), and the LLR calculation follows the MMSE-SIC procedure. PIC avoids SIC ordering and reduces decoding latency because users can be processed in parallel. However, PIC relies directly on the a priori-based soft means of all interfering users, so the quality of the a priori update is especially important. SIC, in contrast, performs ordered stage-by-stage cancellation and can additionally use decoded-user feedback from earlier stages in the same pass. This difference makes PIC and SIC complementary references: PIC represents a more parallel receiver whose cancellation quality is dominated by a priori reliability, whereas SIC represents a sequential receiver that can exploit current decoder outputs at the cost of ordering dependence and longer latency.

## 4.5 MAP-IDD

MAP-IDD provides a coded joint-detection baseline. For user a priori LLRs $L_{a,k,\ell}$, the MAP detector evaluates each candidate vector $\boldsymbol{x}^{(q)} \in \{\pm 1\}^U$ using

$$\Lambda_q = -\frac{\left\| \boldsymbol{y}_\ell - \boldsymbol{H}\boldsymbol{x}^{(q)} \right\|^2}{N_0} + \frac{1}{2}\sum_{k=1}^{U} x_k^{(q)} L_{a,k,\ell}. \tag{4.36}$$

The LLR of user $k$ is

$$L_{k,\ell} = \log \frac{\sum_{q:x_k^{(q)}=+1} \exp(\Lambda_q)}{\sum_{q:x_k^{(q)}=-1} \exp(\Lambda_q)}. \tag{4.37}$$

This receiver provides strong BLER performance but retains exponential complexity in $U$. It is therefore useful as a benchmark for coded MMSE-SIC-IDD and MMSE-PIC-IDD.

## 4.6 Coded Simulation Results

The coded simulations use BPSK modulation, Rayleigh block fading, and a $(1008, 504)$ LDPC code. The receiver candidates include HDS-MAP, MMSE-SIC-IDD with post-

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Figure 4.3: BLER comparison of HDS-MAP, MMSE-SIC-IDD, and MMSE-PIC-IDD for a coded three-user system with a $(1008, 504)$ LDPC code.

SINR ordering, and MMSE-PIC-IDD. Equal large-scale fading $(\beta_1, \ldots, \beta_U) = (1, \ldots, 1)$ and equal transmit power $P_1 = \cdots = P_U = 1$ are used in the coded multiuser comparisons.

The coded results show a different trend from uncoded BER. Although MAP-based detection remains a strong benchmark, MMSE-SIC-IDD becomes much more competitive after channel coding is introduced. Decoder-aided soft cancellation reduces the impact of early SIC errors, and variance-aware filtering prevents the receiver from treating uncertain decisions as perfectly known. This is the key distinction between uncoded SIC and coded SIC-IDD: in the uncoded case, cancellation quality is determined primarily by the detector output at the current SNR, while in the coded case the decoder can use code constraints across the whole block to refine the reliability of the cancellation signal.

For three- and four-user cases, MMSE-SIC-IDD can approach MAP-IDD even with a small number of a priori update iterations. This improvement mainly comes from the inner SIC pass, where decoder outputs of previously detected users are used as soft cancellation terms for the following SIC stages. MAP-IDD still provides favorable BLER

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Figure 4.4: BLER comparison of HDS-MAP, MMSE-SIC-IDD, and MMSE-PIC-IDD for a coded four-user system with a $(1008, 504)$ LDPC code.

performance because it evaluates the coded multiuser likelihood jointly. As the number of users increases, however, MMSE-SIC-IDD becomes more restricted because the residual interference level increases and the ordering problem becomes more difficult. Nevertheless, the coded low-complexity receivers remain attractive because MAP-IDD requires enumeration of all multiuser candidates and therefore becomes rapidly more expensive as $U$ grows.

The coded comparison also shows that for a three-user two-resource coded system, coded MMSE-SIC-IDD can perform close to coded HDS-CDMA with MAP-based detection in the unequal-large-scale-fading case. In the equal-large-scale-fading case, coded HDS-CDMA remains slightly better in the high-SNR region because the joint detector still makes fuller use of the multiuser likelihood. The gain from increasing the number of a priori update iterations is limited under block fading because all coded symbols experience the same channel realization, limiting the amount of independent reliability diversity available to the iterative receiver. This observation suggests that the value of additional IDD iterations depends not only on the code but also on the channel variation across the

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Figure 4.5: BLER comparison of HDS-MAP, MMSE-SIC-IDD, and MMSE-PIC-IDD for a coded six-user system with a $(1008, 504)$ LDPC code.

coded block.

## 4.7 Receiver Tradeoffs

Table 4.1 summarizes the main receiver tradeoffs.

Table 4.1: Qualitative comparison of uplink NOMA receiver families.

| Feature | GDMA/HDS-CDMA | PD-NOMA soft SIC | Multi-resource PD-NOMA MMSE-SIC |
|---|---|---|---|
| User separation | Complex channel gain and/or signature; magnitude and phase | Received-power difference; mainly magnitude | Resource-domain channel vectors; magnitude and phase through filtering |
| Detector type | Joint ML/MAP | Successive cancellation | Linear MMSE filtering plus SIC |
| Complexity | Exponential in number of users | Approximately linear in number of users | Approximately linear to polynomial, depending on matrix inversion size |
| Uncoded BER | Best among compared schemes | Sensitive to error propagation | Intermediate; improved by multiple resources |
| Coded BLER | Strong benchmark | Improved by decoder feedback if IDD is used | Competitive with IDD and variance-aware filtering |
| Latency | Higher due to joint enumeration | Low, but sequential | Low to medium; SIC is sequential, PIC can be parallel |
| System loading | Higher loading possible at higher complexity | Lower loading unless power disparity is strong | Medium loading with lower complexity than MAP |

# Chapter 5 Coded OFDM NOMA

This chapter extends the coded multi-resource receiver to an OFDM uplink. Since practical broadband channels are frequency selective, OFDM is introduced to convert the multipath channel into parallel frequency-domain observations, allowing the MMSE-SIC-IDD framework to be applied on subcarrier or resource-group signals. This chapter studies how channel selectivity, detection ordering, and the choice between SIC and PIC affect BLER.

## 5.1 OFDM System Model

Practical uplink systems often employ OFDM to handle frequency-selective channels. Consider a coded OFDM system with $N_{\text{FFT}}$ subcarriers and cyclic prefix length $N_{\text{CP}}$. Let $N_c$ denote the coded block length and $m$ denote the number of coded bits per modulation symbol. Related coded OFDM and iterative receiver structures have been studied for robust multiuser transmission Lin et al. [2024], Sahin et al. [2024]. Each coded block occupies

$$N_{\text{sym}} = \left\lceil \frac{N_c}{mN_{\text{FFT}}} \right\rceil \tag{5.1}$$

OFDM symbols.

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Figure 5.1: Coded OFDM NOMA system model after CP removal and FFT. Each subcarrier provides a frequency-domain multiuser observation.

After CP removal and FFT, the received signal on subcarrier $q$ of OFDM symbol $j$ is

$$y_{q,j} = \sum_{k=1}^{U} H_{k,q,j} x_{k,q,j} + n_{q,j}. \tag{5.2}$$

For multi-resource NOMA, multiple subcarriers or resource elements can be stacked into a vector model similar to (3.1). Thus, the MMSE-SIC-IDD receiver can be applied subcarrier by subcarrier or blockwise across resource groups.

## 5.2 Multipath Channel Model

The time-domain channel of user $k$ at delay tap $\ell$ is denoted by $h_{k,\ell}$. A five-tap exponentially decayed power-delay profile is considered in the presentation materials. This model is used because it provides a simple way to represent frequency selectivity while keeping the number of parameters manageable. The normalized tap power can be written as

$$\rho_\ell = \frac{e^{-\ell/\lambda}}{\sum_{r=0}^{L_h-1} e^{-r/\lambda}}, \qquad \ell = 0, 1, \ldots, L_h - 1, \tag{5.3}$$

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Figure 5.2: OFDM multipath channel model with exponentially decayed tap powers. The first tap can be modeled as Rician while the remaining taps are Rayleigh.

where $L_h$ is the number of channel taps and $\lambda$ controls the decay rate. Two channel types are considered:

- quasi-static Rayleigh block fading, where all taps are Rayleigh fading paths;
- one-tap Rician plus Rayleigh fading, where the first tap contains a line-of-sight component and the remaining taps are Rayleigh fading paths.

The frequency response on subcarrier $q$ is

$$H_{k,q} = \sum_{\ell=0}^{L_h-1} h_{k,\ell} e^{-j2\pi q\ell/N_{\text{FFT}}}. \tag{5.4}$$

## 5.3 MMSE-SIC-IDD for OFDM

In OFDM NOMA, the detector computes a post-SINR or received-power metric on each subcarrier. Because one LDPC codeword spans many resource elements, the receiver must decide how symbol-level or subcarrier-level reliability should be converted

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Figure 5.3: MMSE-SIC-IDD receiver structure for coded OFDM NOMA. Post-SINR or received-power metrics can be computed over subcarriers to determine the coded-block detection order, and a priori updates refine the soft estimates of all users.

into a coded-block detection order. The detection order of the whole coded block can be determined from an average reliability metric, for example

$$\bar{\Gamma}_k^{\text{post}} = \frac{1}{N_{\text{used}}} \sum_{(q,j)\in\mathcal{D}} \Gamma_{k,q,j}^{\text{post}}, \tag{5.5}$$

where $\mathcal{D}$ is the set of resource elements used by the coded block. A dynamic post-SINR ordering can also be used by recalculating the order after each SIC stage. This may improve ordering accuracy because the remaining interference changes after every cancellation stage, but it increases receiver complexity and control overhead. The comparison between fixed and dynamic ordering is therefore a receiver-design tradeoff rather than a purely performance-driven choice.

## 5.4 OFDM Simulation Settings and Results

The OFDM simulations use BPSK modulation, $U = 3$ users unless otherwise specified, a $(1008, 504)$ LDPC code, $N_{\text{FFT}} = 16$, $N_{\text{CP}} = 4$, perfect CSI, and a five-tap expo-

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Figure 5.4: OFDM BLER comparison between quasi-static Rayleigh and one-tap Rician channel models.

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Figure 5.5: OFDM BLER comparison for different numbers of users.

nentially decayed channel. The compared cases include quasi-static Rayleigh block fading and one-tap Rician fading.

The OFDM results support several observations. First, channel selectivity can provide useful frequency-domain variation, which may help user separation when the receiver exploits subcarrier-dependent channel information. A one-tap Rician component and a quasi-static Rayleigh channel do not simply differ by average power; they also change the distribution of reliability across subcarriers, which affects how much the coded receiver

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Figure 5.6: OFDM BLER comparison of post-SINR ordering and received-power/amplitude-based ordering.

can benefit from soft information. Second, the number of users directly affects the residual interference level and the reliability of SIC cancellation. As the loading increases, the receiver must separate more user vectors from the same resource set, and both ordering errors and residual cancellation errors become more influential.

Third, post-SINR ordering is generally more robust than simple amplitude-based ordering because it accounts for both interference suppression and noise enhancement after MMSE filtering. Fourth, dynamic post-SINR ordering can be useful when the residual channel matrix changes substantially after cancellation, but its additional computation must be justified by the BLER gain. Fifth, MMSE-SIC-IDD and MMSE-PIC-IDD represent a latency-performance tradeoff: SIC can use current-stage decoder outputs, while PIC can process users more parallelly but depends more directly on the a priori soft estimates of all users. Finally, the OMA versus NOMA comparison highlights the core system tradeoff. OMA avoids multiuser interference but consumes more orthogonal resources; NOMA improves loading but requires stronger receiver processing. These results motivate the asynchronous study in the next chapter, where the OFDM receiver must also cope

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Figure 5.7: OFDM BLER comparison with dynamic post-SINR ordering, where the detection order is recalculated after each SIC stage.

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Figure 5.8: OFDM BLER comparison of MMSE-SIC-IDD and MMSE-PIC-IDD.

with user-dependent timing offsets.

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Figure 5.9: OFDM comparison between OMA and NOMA. OMA assigns each user its own resource, while NOMA allows three users to share two resources with MMSE-SIC detection.

# Chapter 6 Asynchronous OFDM NOMA

This chapter studies asynchronous uplink OFDM NOMA, where different users may arrive at the BS with different timing offsets. Although a timing offset within the cyclic prefix can be represented as a subcarrier-dependent phase rotation, the multiuser detector observes different effective channel responses for different users. This chapter examines how timing offsets, MAI/ISI approximations, and detection ordering affect the robustness of MAP and MMSE-SIC receivers.

## 6.1 Motivation

The previous chapters assume perfect symbol-level synchronization among users. In practical uplink random access, users may arrive at the BS with different timing offsets because of propagation delay, oscillator mismatch, or limited timing-advance precision Haci et al. [2017], Lin et al. [2023b]. If the timing drift is within the cyclic prefix, OFDM orthogonality can be largely preserved for a single user, and the drift appears as a subcarrier-dependent phase rotation in the frequency domain. For multiuser NOMA, however, different users may have different timing offsets, which changes their effec-

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Figure 6.1: Asynchronous uplink OFDM NOMA model. Timing offsets within the cyclic prefix appear as user-dependent phase rotations across subcarriers.

tive channel coefficients and may introduce multiple-access interference (MAI) or inter-symbol interference (ISI) depending on the receiver alignment.

## 6.2 Timing Offset Model

Let $\tau_k$ denote the timing offset of user $k$ relative to the receiver timing reference. If $\tau_k$ is within the CP tolerance, the frequency-domain effective channel on subcarrier $q$ can be written as

$$\tilde{H}_{k,q} = H_{k,q} e^{-j2\pi q \tau_k / N_{\text{FFT}}}. \tag{6.1}$$

The received signal becomes

$$y_q = \sum_{k=1}^{U} \tilde{H}_{k,q} x_{k,q} + \eta_q, \tag{6.2}$$

where $\eta_q$ includes AWGN and possible residual interference terms caused by imperfect alignment. If the receiver is aligned to user 1, the BLER of user 1 can be evaluated under different offsets of the other users.

The presentation materials consider system labels such as $(2,3,0,3)$ and $(2,4,0,4)$, which can be interpreted as two resources, three or four users, and timing offsets selected over a specified range. This notation is convenient for comparing how loading and timing spread affect the receiver. It also emphasizes that asynchronous behavior should not be treated only as a synchronization impairment; it changes the effective channel geometry and may therefore alter the relative advantage of joint detection and linear SIC detection.

## 6.3 MAI and ISI Variance Approximation

When timing offsets or excess delays cause part of a multipath component to fall outside the useful FFT window, residual ISI can be approximated as additional noise. Let $\Delta_\ell$ denote the excess delay of tap $\ell$ beyond the receiver-aligned useful interval. For an exponentially decayed $L_h$-tap channel, the normalized tap power is given by (5.3). An approximate ISI power ratio for user $k$ can be written as

$$\alpha_k^{\mathrm{ISI}} = \sum_{\ell \in \mathcal{L}_{\mathrm{out}}} \rho_\ell, \tag{6.3}$$

where $\mathcal{L}_{\mathrm{out}}$ is the set of taps whose delayed energy is outside the useful window. If the ISI component is modeled as additional Gaussian noise, the effective SINR is approximated by

$$\Gamma_k^{\mathrm{eff}} = \frac{P_k \left|\tilde{H}_{k,q}\right|^2}{N_0 + \sigma_{\mathrm{MAI}}^2 + \sigma_{\mathrm{ISI},k}^2}. \tag{6.4}$$

This approximation is especially useful for APP or MAP-style detectors, where the additional interference variance can be included in the likelihood metric. It is also useful for interpreting MMSE-SIC results, because additional MAI/ISI variance reduces the reliability of both ordering and cancellation. When the additional interference dominates,

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Figure 6.2: MAI/ISI variance approximation for asynchronous OFDM NOMA. Timing offsets within or near the CP modify the effective channel and can be modeled as additional interference variance.

the difference between refined post-SINR ordering and simpler amplitude-based ordering may become less visible.

## 6.4 Asynchronous Simulation Results

The asynchronous simulations evaluate BLER under single-resource and two-resource settings. The receiver may be aligned to user $1$ to observe the BLER of that user. Multi-resource systems such as $(2, 3, 0, 3)$ and $(2, 4, 0, 4)$ are considered to compare MAP detection and MMSE-SIC under different timing-offset ranges.

The main conclusion from the asynchronous study is that MMSE-SIC can be slightly better than MAP detection in some two-resource three-user or four-user asynchronous settings. This is notable because in the corresponding synchronous four-user system, MAP detection can significantly outperform MMSE-SIC. The contrast suggests that asynchronous phase rotations and MAI change the effective separability of the users. A receiver that is optimal for the ideal synchronous likelihood is not necessarily the most robust

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Figure 6.3: Asynchronous single-resource BLER result with the receiver aligned to user 1.

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Figure 6.4: Asynchronous BLER result for the $(2, 3, 0, 3)$ system, case A.

option when the practical impairment model changes the effective observation statistics.

MMSE-SIC, which relies on linear filtering and successive cancellation over the effective channel, can exploit part of this variation and may be more robust to certain asynchronous conditions. The BLER of post-SINR ordering and amplitude-based ordering becomes nearly identical in some asynchronous cases, particularly when MAI is present and dominates the ordering difference. This result does not imply that ordering is unimportant in general; rather, it indicates that the benefit of a refined ordering metric depends

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Figure 6.5: Asynchronous BLER result for the $(2, 3, 0, 3)$ system, case B.

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Figure 6.6: Asynchronous BLER result for the $(2, 3, 0, 3)$ system, case C.

on whether the receiver is limited mainly by multiuser separability or by additional asynchronous interference. The asynchronous results therefore broaden the thesis comparison from idealized detector performance to receiver robustness under a more realistic uplink impairment.

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Figure 6.7: Asynchronous BLER result for the $(2, 4, 0, 4)$ system, case A.

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Figure 6.8: Asynchronous BLER result for the $(2, 4, 0, 4)$ system, case B.

# Chapter 7 Conclusion and Future Work

This chapter summarizes the main results of this thesis and discusses possible future work. The conclusions are organized around the receiver-side factors studied in the previous chapters, including channel information, decoder reliability, frequency-domain diversity, and synchronization conditions in uplink NOMA systems.

## 7.1 Conclusion

This thesis draft compared uplink NOMA receivers from uncoded single-resource systems to coded multi-resource and OFDM systems. In the single-resource case, GDMA outperforms PD-NOMA with soft SIC because GDMA jointly exploits complex channel gains, while SIC depends heavily on received-power ordering and suffers from error propagation. Increasing the large-scale fading disparity helps SIC, but it also creates user-fairness concerns.

In the uncoded multi-resource case, MMSE-SIC improves PD-NOMA by using vector observations across resources. Post-SINR ordering is more effective than amplitude-based ordering because it reflects post-filtering interference and noise enhancement. Nevertheless, HDS-CDMA with ML detection still achieves better BER because it performs joint multiuser detection.

In coded systems, MMSE-SIC-IDD becomes much more competitive. Decoder-aided soft cancellation reduces SIC error propagation, and variance-aware MMSE filtering uses decoder reliability to avoid overconfident cancellation. MMSE-PIC-IDD and MAP-IDD provide useful comparisons: PIC reduces sequential latency, while MAP offers a high-performance but high-complexity benchmark. The coded results show that receiver architecture and channel coding should be considered jointly rather than separately, because decoder feedback changes the quality of the cancellation signal.

For coded OFDM NOMA, channel selectivity, detection ordering, and the choice between SIC and PIC all affect BLER. The OFDM extension shows that the same MMSE-SIC-IDD principle can be applied to frequency-domain observations, but coded-block ordering and subcarrier reliability become additional design issues. In asynchronous OFDM NOMA, timing offsets within the CP introduce effective channel phase rotations and possible additional interference; under some settings, MMSE-SIC shows stronger robustness than expected from synchronous comparisons. Overall, the thesis suggests that low-complexity SIC-based receivers can be made competitive when they are supported by multiple resources, soft information, variance-aware filtering, and receiver designs that reflect the actual uplink impairment model.

## 7.2 Future Work

Several research directions remain open:

- Extend the simulation study to more users, more resources, and higher-order modulation such as QPSK or 16-QAM.
- Study fast-fading coded blocks, where a priori update iterations may provide larger gains than in block-fading channels.
- Develop lower-complexity approximations to MAP-IDD for HDS-CDMA and compare them with MMSE-SIC-IDD under the same latency constraints.
- Analyze imperfect CSI more deeply, including its effect on SIC ordering, cancellation accuracy, and variance-aware filtering.
- Optimize user pairing and resource assignment to balance SIC performance and user fairness.
- Extend the asynchronous OFDM study to random access scenarios with unknown timing offsets and blind or semi-blind timing estimation.

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